

# Project Methodology Summary Report

## 1. Data Loading

- **Datasets:** train.csv, public\_test.csv, and private\_test.csv were loaded into pandas DataFrames. These datasets contain user interaction data, with train.csv including a Converted target variable.

## 2. Preprocessing and Feature Engineering

- **Target and Feature Separation:** The Converted column was identified as the target variable (y\_train).
- **User ID Handling:** User\_ID was explicitly identified as an identifier and excluded from the feature set for model training. User\_IDs from private\_test were stored separately for submission.
- **Categorical Feature Encoding:** Categorical features (Device\_Type, Traffic\_Source) were transformed using one-hot encoding to convert them into a numerical format suitable for the model. This was done by concatenating train and private test sets to ensure consistent encoding across both datasets before splitting them back.
- **Missing Value Imputation:** Missing numerical values in the feature sets were handled using SimpleImputer with a median strategy. The imputer was fitted on the training data and then used to transform both training and private test data.
- **Feature Scaling:** Features were scaled using StandardScaler to achieve zero mean and unit variance. The scaler was fitted on the imputed training data and applied to both training and private test data to prevent data leakage.
- **Class Imbalance Handling:** The target variable Converted was found to have an imbalance (more '0's than '1's). To address this, scale\_pos\_weight was calculated based on the ratio of negative to positive classes and applied during model training.

## 3. Model Training

- **Model Selection:** An XGBClassifier was chosen for its robust performance in classification tasks, replacing an initial LogisticRegression model.
- **Hyperparameters:** The XGBClassifier was initialized with random\_state=42 for reproducibility, eval\_metric='logloss', and critically, the calculated scale\_pos\_weight to mitigate class imbalance.
- **Training:** The model was trained on the preprocessed and scaled training features (X\_train\_scaled) and the target variable (y\_train).

## 4. Prediction and Submission

- **Prediction Generation:** The trained XGBoost model was used to generate predictions on the X\_private\_scaled dataset.
- **Submission File Creation:** The predictions were combined with the original User\_IDs from the private test set into a pandas DataFrame.
- **Output:** The final submission DataFrame was saved to submission.csv without the index, in the required format for competition submission.